Question 1: File Permissions Audit

During an audit of a Unix system, you encounter a file with the following permissions: `-rwxr-xr--`.

- Calculate the numeric (octal) representation of these permissions.
- Determine the total number of users who have write access to this file.

Solution:

1. **Numeric Representation:**
   - Owner (rwx): Read (4) + Write (2) + Execute (1) = 7
   - Group (r-x): Read (4) + Execute (1) = 5
   - Others (r--): Read (4) = 4

   Therefore, the numeric representation is `754`.
2. **Users with Write Access:**
   - Owner has write permission.
   - Group and Others do not have write permission.

   Assuming the system has 100 users, with 1 owner and 99 others, only the owner has write access.
   Total users with write access: **1**

Question 2: Password Policy Compliance

An organization's security policy mandates that user passwords must be changed every 90 days. During an audit, you find that out of 150 user accounts, 45 have not changed their passwords in the last 120 days.

- Calculate the percentage of user accounts that are non-compliant with the password policy.

Solution:

Non-compliant accounts: 45
Total accounts: 150
Percentage of non-compliance = (Non-compliant accounts / Total accounts) × 100
= (45 / 150) × 100
= 0.3 × 100
= **30%**

Question 3: Disk Usage Analysis

During a Unix system audit, you analyze disk usage and find the following:

- Total disk space: 500 GB
- Used disk space: 275 GB
- Calculate the percentage of disk space currently in use.
- If the organization's policy requires a maximum of 60% disk usage, determine the amount of data (in GB) that needs to be cleared to comply with this policy.

Solution:

1. **Percentage of Disk Space Used:**
   Percentage used = (Used disk space / Total disk space) × 100
   = (275 GB / 500 GB) × 100
   = 0.55 × 100
   = **55%**
   The current disk usage is 55%, which is within the 60% limit.
2. **Data to Clear:**
   No data needs to be cleared as the current usage is within the policy limit.

Question 4: User Account Management

An audit reveals that a Unix system has 200 user accounts. Of these, 50 accounts have never been logged into, and 30 accounts have been inactive for over a year.

- Calculate the percentage of accounts that are either unused or inactive.
- Determine the total number of active accounts.

Solution:

1. **Percentage of Unused or Inactive Accounts:**
   Unused accounts: 50
   Inactive accounts: 30

Total unused or inactive accounts = 50 + 30 = 80
Percentage = (80 / 200) × 100
= 0.4 × 100
= **40%**

2. **Total Active Accounts:**
   Active accounts = Total accounts - Unused or inactive accounts
   = 200 - 80
   = **120**

Question 5: Log File Monitoring

During an audit, you observe that log files are generated at an average rate of 2 GB per day. The current log storage has a capacity of 100 GB, with 40 GB already in use.

- Calculate the number of days remaining before the log storage reaches full capacity.
- If the organization plans to implement a log rotation policy that deletes logs older than 30 days, determine the maximum average daily log generation rate (in GB) that would keep the storage usage at or below 100 GB.

Solution:

1. **Days Remaining Before Full Capacity:**
   Available storage = Total capacity - Used capacity
   = 100 GB - 40 GB
   = 60 GB
   Days remaining = Available storage / Daily log generation rate
   = 60 GB / 2 GB per day
   = **30 days**
2. **Maximum Average Daily Log Generation Rate with 30-Day Retention:**
   To keep storage usage within 100 GB over a 30-day period:
   Maximum total log size = 100 GB
   Maximum daily log generation rate = Maximum total log size / Retention period
   = 100 GB / 30 days

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1. Auditing Account Management and Password Controls

Scenario:

A company has **500 user accounts** on its Unix system. A recent audit revealed the following:

- **30% of accounts** have weak passwords that do not comply with the organization's password policy.
- **15% of accounts** are inactive (no login activity in the past 180 days).
- **5% of accounts** are shared accounts (used by multiple users, violating policy).

The security team decides to disable all non-compliant accounts and enforce stricter password policies.

Questions:

1. **How many accounts will be disabled due to non-compliance?**
2. **If disabling an account takes an average of 2 minutes per account, how long (in hours) will it take to disable all non-compliant accounts?**

Solution:

1. **Total Non-Compliant Accounts:**
   - Weak passwords: **30% of 500** = 30/100×500=150
   - Inactive accounts: **15% of 500** = 15/100×500=75
   - Shared accounts: **5% of 500** = 5/100×500=25

   Total Accounts to be Disabled = 150 + 75 + 25 = **250 accounts**
2. **Time Required to Disable Accounts:**
   - Time per account = **2 minutes**
   - Total time = 250×2 minutes = **500 minutes**
   - Convert to hours: 500/60=8.33 hours

Final Answer: 250 accounts will be disabled, taking approximately 8 hours 20 minutes.

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2. Auditing File Security and Controls

Scenario:

A Unix file server contains **1,000,000 files**, distributed as follows:

- **20% are owned by inactive or disabled users.**
- **10% have incorrect permission settings that violate security policies.**
- **5% are world-writable (i.e., open to anyone).**

The IT department needs to identify and secure these files.

Questions:

1. **How many files need corrective action?**
2. **If scanning each file for security compliance takes 0.005 seconds, how long (in hours) will the full audit take?**

Solution:

1. **Files Requiring Corrective Action:**
   - Inactive user files = 20%×1,000,000=200,000
   - Incorrect permissions = 10%×1,000,000=100,000
   - World-writable files = 5%×1,000,000=50,000

   Total affected files = 200,000 + 100,000 + 50,000 = **350,000 files**
2. **Time Required for Full Audit:**
   - Time per file = **0.005 seconds**
   - Total time = 350,000×0.005 seconds = **1,750 seconds**
   - Convert to hours: 1,750/3600=0.486 hours

   Final Answer: 350,000 files require corrective action, and the full audit will take approximately 29 minutes.

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3. Auditing Network Security and Controls

Scenario:

A Unix-based organization has **200 servers**, each running **30 network services** on average. An audit finds:

- **15% of services are outdated and vulnerable to attacks.**
- **5% of services have open ports that should be closed.**
- **10% of services are running with root privileges unnecessarily.**

The IT security team wants to mitigate these issues by updating, reconfiguring, or disabling the affected services.

Questions:

1. **How many services in total are affected?**
2. **If fixing each issue takes 3 minutes, how long (in hours) will the remediation take?**

Solution:

1. **Total Network Services:**
   - **Total Services =** 200×30=6000

   Affected Services:
   - Outdated: 15%×6000=900
   - Open Ports: 5%×6000=300
   - Root Privilege Issues: 10%×6000=600

   Total Affected Services = 900 + 300 + 600 = 1,800 services
2. **Total Remediation Time:**
   - Time per service = **3 minutes**
   - Total time = 1800×3 minutes = **5,400 minutes**
   - Convert to hours: 5400/60=90 hours

Final Answer: 1,800 services need remediation, requiring 90 hours to complete.

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4. Auditing Audit Logs

Scenario:

A Linux system generates **500MB of logs per day**. An audit reveals:

- **40% of logs contain security-related events.**
- **25% of logs need to be analyzed for potential threats.**
- The system retains logs for **90 days** before archiving.

The security team needs to analyze logs efficiently while ensuring enough storage capacity.

Questions:

1. **What is the total size of logs stored before archiving?**
2. **If analyzing 1MB of logs takes 0.2 seconds, how long (in hours) will it take to analyze all logs marked for review?**

Solution:

1. **Total Stored Logs Before Archiving:**
   - Logs per day = **500MB**
   - Retention period = **90 days**
   - **Total log size =** 500×90 MB = **45,000 MB (45 GB)**
2. **Time Required for Log Analysis:**
   - Logs for review = 25%×45,000= 11,250MB
   - Time per MB = **0.2 seconds**
   - Total time = 11,250×0.2 seconds = **2,250 seconds**
   - Convert to hours: 2,250/3600=0.625 hours

   Final Answer: The system stores 45GB of logs, and reviewing critical logs will take approximately 38 minutes.

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5. Auditing System Resource Utilization

Scenario:

A company runs **50 Unix servers**, each with:

- **64GB of RAM**
- **500GB disk space**
- **8-core processors**

The audit finds:

- **25% of RAM is consistently unused.**
- **15% of disk space is filled with temporary files.**
- **20% of CPU cores remain idle.**

Questions:

1. **How much total RAM across all servers is wasted?**
2. **What is the total disk space occupied by temporary files?**

Solution:

1. **Total Unused RAM:**
   - RAM per server = **64GB**
   - Total RAM = 50×64 GB = **3,200 GB**
   - Wasted RAM = 25%×3200=800GB
2. **Disk Space Occupied by Temporary Files:**
   - Disk per server = **500GB**
   - Total Disk Space = 50×500=25,000GB
   - Temp files = 15%×25,000=3,750GB

   Final Answer: 800GB of RAM is wasted, and 3.75TB of disk space is occupied by temporary files.